

Week 2: Making Decisions & Repeating Tasks in Python

1. Create a program that gives different messages based on time of day
2. Build a number guessing game using if-else statements
3. Use for loop to print numbers from 1 to 100
4. Create a function that calculates area of rectangle
5. Make a simple calculator that can add, subtract, multiply, divide
6. Practice with real-life scenarios: grading system, age categories

Project:

Build a Simple Student Grade Calculator that takes marks and returns grade (A, B, C, etc.) with encouraging messages for each grade

Tips & Resources:

- Search YouTube: 'Python if else statements for beginners'
- Start with simple conditions and gradually make them complex
- Use paper and pen to write logic before coding
- Test each small part before moving to next
- Celebrate small wins - each working program is an achievement

1. Create a program that gives different messages based on time of day

```
In [1]: from datetime import datetime

# Get current hour
now = datetime.now()
hour = now.hour

print("Current Time:", now.strftime("%I:%M %p"))
# Decide message based on time
if 5 <= hour < 12:
    print("Good Morning! Have a wonderful day ahead ☀️")
elif 12 <= hour < 16:
    print("Good Afternoon! Hope your day is going well 🌸")
elif 16 <= hour < 20:
    print("Good Evening! Relax and unwind ✨")
else:
    print("Good Night! Time to rest and recharge 🌙")
```

Current Time: 08:14 AM

Good Morning! Have a wonderful day ahead ☀️

2. Build a number guessing game using if-else statements

```
In [2]: import random

secret = random.randint(1,100)

while True:
    guess = int(input("Guess a number:"))
    if guess == secret:
        print("Correct! You guessed it right!")
        break
    elif guess < secret:
        print("Guess Higher!")
    else:
        print("Guess Lower!")
```

Guess Higher!
Guess Lower!
Guess Lower!
Guess Higher!
Guess Lower!
Correct! You guessed it right!

3. Use for loop to print numbers from 1 to 100

```
In [3]: for i in range(1, 101):
        print(i ,end=" - ")
```

1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10 - 11 - 12 - 13 - 14 - 15 - 16 - 17 - 18 - 19 - 20 -
21 - 22 - 23 - 24 - 25 - 26 - 27 - 28 - 29 - 30 - 31 - 32 - 33 - 34 - 35 - 36 - 37 - 38 - 3
9 - 40 - 41 - 42 - 43 - 44 - 45 - 46 - 47 - 48 - 49 - 50 - 51 - 52 - 53 - 54 - 55 - 56 - 57
- 58 - 59 - 60 - 61 - 62 - 63 - 64 - 65 - 66 - 67 - 68 - 69 - 70 - 71 - 72 - 73 - 74 - 75 -
76 - 77 - 78 - 79 - 80 - 81 - 82 - 83 - 84 - 85 - 86 - 87 - 88 - 89 - 90 - 91 - 92 - 93 - 9
4 - 95 - 96 - 97 - 98 - 99 - 100 -

4. Create a function that calculates area of rectangle

```
In [4]: l=5
b=2
def area_rectangle(l,b):
    if l <= 0 or b <= 0:
        return "Invalid dimensions! Length and breadth must be positive."
    return l*b

print(f"The area of rectangle with length {l} & breadth {b} is {area_rectangle(l,b)}.")
```

The area of rectangle with length 5 & breadth 2 is 10

5. Make a simple calculator that can add, subtract, multiply, divide

```
In [5]: print("Welcome to the Chained Calculator!")
print("You can keep using the result of the previous calculation.")
print("\nOperations:")
print("1. Addition (+)")
print("2. Subtraction (-)")
print("3. Multiplication (*)")
print("4. Division (/)")
print("5. Exit\n")

first_time = True # To check if it's the first calculation

while True:
    if first_time:
        num1 = float(input("Enter first number: "))
        first_time = False
    else:
        num1 = result
        print(f"\nFirst number (from previous result): {num1}")

    num2 = float(input("Enter next number: "))
    choice = input("Enter operation (1/2/3/4/5 or +, -, *, /): ")

    if choice == '1' or choice == '+':
        result = num1 + num2
        print(f"Result: {num1} + {num2} = {result}")
    elif choice == '2' or choice == '-':
        result = num1 - num2
        print(f"Result: {num1} - {num2} = {result}")
    elif choice == '3' or choice == '*':
        result = num1 * num2
        print(f"Result: {num1} * {num2} = {result}")
    elif choice == '4' or choice == '/':
        if num2 == 0:
            print("Error! Division by zero is not allowed.")
            continue
        else:
            result = num1 / num2
            print(f"Result: {num1} / {num2} = {result}")
    elif choice == '5':
        print("Exiting calculator. Goodbye!")
        break
    else:
        print("Invalid input! Please select a valid operation.")
        continue
```

Welcome to the Chained Calculator!

You can keep using the result of the previous calculation.

Operations:

1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Exit

Result: $80.0 + 20.0 = 100.0$

First number (from previous result): 100.0

Result: $100.0 - 50.0 = 50.0$

First number (from previous result): 50.0

Result: $50.0 * 4.0 = 200.0$

First number (from previous result): 200.0

Result: $200.0 / 5.0 = 40.0$

First number (from previous result): 40.0

Exiting calculator. Goodbye!